Discrete Structures(CT-162)

Batch 2024 and Back Loggers

Fall 2024

Assignment No. 1

Book:

Discrete Mathematics and Its Applications (8th Edition)

By Kenneth H. Rosen

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Q.5,7,9,10,17,18,25,26,35,36,44,45

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Q.4,8,9,10,12,14,18,36

Convert 180_{10} into a binary number.

Convert 56_{10} into a binary number.

Convert the binary number 0101001_2 into a decimal system.

Convert the given binary number 100001_2 into a decimal number system.

Subtract (1010)2(1010)2 from (1111)2(1111)2 using 2's complement method.

Subtract (1010)2(1010)2 from (1000)2(1000)2 using 2's complement.

Perform (1010) - (101111) using 1's and 2's complement.

Subtract (1010) from (1111) using 1's and 2's complement.

Add the following binary numbers:

(i) 11110000 + 1000011001

(ii) 10101000 + 010001001

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5. Use a truth table to verify the distributive law $p \land (q \lor r) \equiv (p \land q) \lor (p \land r)$.

7. Use De Morgan's laws to find the negation of each of the following statements.
a) Jan is rich and happy.
b) Carlos will bicycle or run tomorrow.
c) Mei walks or takes the bus to class.
d) Ibrahim is smart and hard working.

9. For each of these compound propositions, use the conditional-disjunction equivalence (Example 3) to find an equivalent compound proposition that does not involve conditionals.
a) $p \rightarrow \neg q$
b) $(p \rightarrow q) \rightarrow r$
c) $(\neg q \rightarrow p) \rightarrow (p \rightarrow \neg q)$

10. For each of these compound propositions, use the conditional-disjunction equivalence (Example 3) to find an equivalent compound proposition that does not involve conditionals.
a) $\neg p \rightarrow \neg q$
b) $(p \lor q) \rightarrow \neg p$
c) $(p \rightarrow \neg q) \rightarrow (\neg p \rightarrow q)$

17. Use truth tables to verify the absorption laws.
a) $p \lor (p \land q) \equiv p$
b) $p \land (p \lor q) \equiv p$

18. Determine whether $(\neg p \land (p \to q)) \to \neg q$ is a tautology.

25. Show that $\neg(p \leftrightarrow q)$ and $\neg p \leftrightarrow q$ are logically equivalent.

26. Show that $(p \to q) \land (p \to r)$ and $p \to (q \land r)$ are logically equivalent.

35. Show that $(p \to q) \to r$ and $p \to (q \to r)$ are not logically equivalent.

36. Show that $(p \land q) \to r$ and $(p \to r) \land (q \to r)$ are not logically equivalent.

44. Find a compound proposition involving the propositional variables p, q, and r that is true when p and q are true and r is false, but is false otherwise. [Hint: Use a conjunction of each propositional variable or its negation.]

45. Find a compound proposition involving the propositional variables p, q, and r that is true when exactly two of p, q, and r are true and is false otherwise. [Hint: Form a disjunction of conjunctions. Include a conjunction for each combination of values for which the compound proposition is true. Each conjunction should include each of the three propositional variables or its negations.]

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4. Let $P(x, y)$ be the statement "Student x has taken class y," where the domain for x consists of all students in your class and for y consists of all computer science courses at your school. Express each of these quantifications in English.

a) $\exists x \exists y P(x, y)$
b) $\exists x \forall y P(x, y)$
c) $\forall x \exists y P(x, y)$
d) $\exists y \forall x P(x, y)$
e) $\forall y \exists x P(x, y)$
f) $\forall x \forall y P(x, y)$

8. Let $Q(x, y)$ be the statement "Student x has been a contestant on quiz show y." Express each of these sentences in terms of $Q(x, y)$, quantifiers, and logical connectives, where the domain for x consists of all students at your school and for y consists of all quiz shows on television.

a) There is a student at your school who has been a contestant on a television quiz show.
b) No student at your school has ever been a contestant on a television quiz show.
c) There is a student at your school who has been a contestant on Jeopardy! and on Wheel of Fortune.
d) Every television quiz show has had a student from your school as a contestant.
e) At least two students from your school have been contestants on Jeopardy!.

9. Let L(x, y) be the statement "x loves y," where the domain for both x and y consists of all people in the world. Use quantifiers to express each of these statements.
a) Everybody loves Jerry.
b) Everybody loves somebody.
c) There is somebody whom everybody loves.
d) Nobody loves everybody.
e) There is somebody whom Lydia does not love.
f) There is somebody whom no one loves.
g) There is exactly one person whom everybody loves.
h) There are exactly two people whom Lynn loves.
i) Everyone loves himself or herself.
j) There is someone who loves no one besides himself or herself.

10. Let F(x, y) be the statement "x can fool y," where the domain consists of all people in the world. Use quantifiers to express each of these statements.
a) Everybody can fool Fred.
b) Evelyn can fool everybody.
c) Everybody can fool somebody.
d) There is no one who can fool everybody.
e) Everyone can be fooled by somebody.
f) No one can fool both Fred and Jerry.
g) Nancy can fool exactly two people.
h) There is exactly one person whom everybody can fool.
i) No one can fool himself or herself.
j) There is someone who can fool exactly one person besides himself or herself

12. Let I(x) be the statement "x has an Internet connection" and C(x, y) be the statement "x and y have chatted over the Internet," where the domain for the variables x and y consists of all students in your class. Use quantifiers to express each of these statements.

a) Jerry does not have an Internet connection.
b) Rachel has not chatted over the Internet with Chelsea.
c) Jan and Sharon have never chatted over the Internet.
d) No one in the class has chatted with Bob.
e) Sanjay has chatted with everyone except Joseph.
f) Someone in your class does not have an Internet connection.
g) Not everyone in your class has an Internet connection.
h) Exactly one student in your class has an Internet connection.
i) Everyone except one student in your class has an Internet connection.
j) Everyone in your class with an Internet connection has chatted over the Internet with at least one other student in your class.
k) Someone in your class has an Internet connection but has not chatted with anyone else in your class.
l) There are two students in your class who have not chatted with each other over the Internet.
m) There is a student in your class who has chatted with everyone in your class over the Internet.
n) There are at least two students in your class who have not chatted with the same person in your class.
o) There are two students in the class who between them have chatted with everyone else in the class.

14. Use quantifiers and predicates with more than one variable to express these statements.

a) There is a student in this class who can speak Hindi.
b) Every student in this class plays some sport.
c) Some student in this class has visited Alaska but has not visited Hawaii.
d) All students in this class have learned at least one programming language.
e) There is a student in this class who has taken every course offered by one of the departments in this school.
f) Some student in this class grew up in the same town as exactly one other student in this class.
g) Every student in this class has chatted with at least one other student in at least one chat group

18. Express each of these system specifications using predicates, quantifiers, and logical connectives, if necessary.

a) At least one console must be accessible during every fault condition.

b) The e-mail address of every user can be retrieved whenever the archive contains at least one message sent by every user on the system.

c) For every security breach there is at least one mechanism that can detect that breach if and only if there is a process that has not been compromised.

d) There are at least two paths connecting every two distinct endpoints on the network.

e) No one knows the password of every user on the system except for the system administrator, who knows all passwords

36. Express each of these statements using quantifiers. Then form the negation of the statement so that no negation is to the left of a quantifier. Next, express the negation in simple English. (Do not simply use the phrase "It is not the case that.")

a) No one has lost more than one thousand dollars playing the lottery.

b) There is a student in this class who has chatted with exactly one other student. 72 1

c) No student in this class has sent e-mail to exactly two other students in this class.

d) Some student has solved every exercise in this book.

e) No student has solved at least one exercise in every section of this book.

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Question # 5

The two propositions $p \wedge (q \vee r)$ and $(p \wedge q) \vee (p \wedge r)$ are equivalent because the last two columns of Truth Table are identical.

p	q	r	$p \wedge q$	$p \wedge r$	$q \vee r$	$p \wedge (q \vee r)$	$(p \wedge q) \vee (p \wedge r)$
T	T	T	T	T	T	T	T
T	T	F	T	F	T	T	T
T	F	T	F	T	T	T	T
T	F	F	F	F	F	F	F
F	T	T	F	F	T	F	F
F	T	F	F	F	T	F	F
F	F	T	F	F	T	F	F
F	F	F	F	F	F	F	F

By constructing truth table it is proved that the $p \wedge (q \vee r)$ is logically equivalent to $(p \wedge q) \vee (p \wedge r)$ for any truth value of p, q, and r.

QUESTION # 7

De Morgan's Law

$\neg(p \wedge q) \equiv \neg p \vee \neg q$
$\neg(p \vee q) \equiv \neg p \wedge \neg q$

a ℞

Jan is not rich or not happy

b ℞

Carlos will not bycycle and not run tomorrow.

c ℞

Mei does not walk and does not take the bus to class

d ℞

Ibrahim is not smart or not hard working.

QUESTION # 9

Conditional disjunction equivalence

$$p \to q \equiv \neg p \vee q$$

De Morgan's Law

$$\neg(p \wedge q) \equiv \neg p \vee \neg q$$
$$\neg(p \vee q) \equiv \neg p \wedge \neg q$$

Double negation Law

$$\neg(\neg p) \equiv p$$

a

$p \to \neg q$

$\equiv \neg p \vee \neg q$ conditional disjunction equivalence

$\equiv \neg(p \wedge q)$ De Morgan's Law

b

$(p \to q) \to r$

$\equiv (\neg p \vee q) \to r$ conditional disjunction eq.

$\equiv \neg(\neg p \vee q) \vee r$ conditional disjunction eq

$\equiv (\neg(\neg p) \wedge \neg q) \vee r$ De Morgan's Law

Date:

$\equiv (p \wedge \neg q) \vee r$ Double negation law

c

$(\neg q \rightarrow p) \rightarrow (p \rightarrow \neg q)$

$\equiv (\neg(\neg q) \vee p) \rightarrow (\neg p \vee \neg q)$ conditional dis. eq

$\equiv q \vee p \rightarrow (\neg p \vee \neg q)$ double negation

$\equiv \neg(q \vee p) \vee (\neg p \vee \neg q)$ conditional disjunction eq

$\equiv \neg(q \vee p) \vee \neg(p \wedge q)$ De Morgan's Law

$\equiv \neg((q \vee p) \wedge (p \wedge q))$ De Morgan's Law

Hence, the condition is satisfied, it does not contain conditionals anymore.

QUESTION #10

Conditional disjunction equivalence

$$p \rightarrow q \equiv \neg p \vee q$$

Double negation law

$$\neg(\neg p) \equiv p$$

Date:

De Morgan's Law

$$\neg(p \wedge q) \equiv \neg p \vee \neg q$$
$$\neg(p \vee q) \equiv \neg p \wedge \neg q$$

Absorption Laws

$$p \vee (p \wedge q) \equiv p$$
$$p \wedge (p \vee q) \equiv p$$

a

$\neg p \rightarrow \neg q$

$\equiv \neg(\neg p) \vee \neg q$ conditional disjunction eq.

$\equiv p \vee \neg q$ double negation law

b

$(p \vee q) \rightarrow \neg p$

$\equiv \neg(p \vee q) \vee \neg p$ conditional disjunction eq.

$\equiv (\neg p \wedge \neg q) \vee \neg p$ De Morgan's Law

$\equiv \neg p$ Absorption Law

c

$(p \rightarrow \neg q) \rightarrow (\neg p \rightarrow q)$

$\equiv (\neg p \vee \neg q) \rightarrow (\neg(\neg p) \vee q)$ conditional disjunction eq.

$\equiv (\neg p \vee \neg q) \rightarrow (p \vee q)$ double negation

$\equiv \neg(\neg p \vee \neg q) \vee (p \vee q)$ conditional disjunction eq.

$\equiv (\neg(\neg p) \wedge \neg(\neg q)) \vee (p \vee q)$ De Morgan's law

$\equiv (p \wedge q) \vee (p \vee q)$ double negation

QUESTION #17

a b

The two proposition $p \vee (p \wedge q)$ and p equivalent because the last two columns of following truth table are identical.

p	q	$p \wedge q$	$p \vee (p \wedge q)$	p
T	T	T	T	T
T	F	F	T	T
F	T	F	F	F
F	F	F	F	F

Hence verified.

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b 18

The two propositions $p \wedge (q \vee p)$ and and p are equivalent because the last two columns of truth table are identicle.

p	q	$p \vee q$	$p \wedge (p \vee q)$	p
T	T	T	T	T
T	F	T	T	T
F	T	T	F	F
F	F	F	F	F

Hence, verified.

QUESTION # 18

$(\neg p \wedge (p \to q)) \to \neg q$

p	q	$p \to q$	$\neg p$	$(\neg p \wedge (p \to q))$	$\neg q$	$(\neg p \wedge (p \to q)) \to \neg q$
T	T	T	F	F	F	T
T	F	F	F	F	T	T
F	T	T	T	T	F	F
F	F	T	T	T	T	T

The conditional statement is not a tautology, because the last column in the truth table contains a False; a tautology would contain only true values.

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QUESTION #25

Logical equivalence

$$p \to q \equiv \neg p \lor q \quad (1)$$

$$p \leftrightarrow q \equiv (p \to q) \land (q \to p) \quad (2)$$

$\neg(p \leftrightarrow q) \equiv \neg p \leftrightarrow q$

Take $\neg(p \leftrightarrow q)$

$\equiv \neg[(p \to q) \land (q \to p)]$ using logical eq; (2)

$\equiv \neg(p \to q) \lor \neg(q \to p)$ De Morgan's law

$\equiv \neg(\neg p \lor q) \lor \neg(\neg q \lor p)$ logical eq; (1)

$\equiv (p \land \neg q) \lor (q \land \neg p)$ De Morgan & Double neg

$\equiv ((p \land \neg q) \lor q) \land ((p \land \neg q) \lor \neg p))$ distributive

$\equiv ((p \lor q) \land (\neg q \lor q)) \land ((p \lor \neg p) \land (\neg q \lor \neg p))$ distributive

$\equiv ((p \lor q) \land T) \land (T \land (\neg q \lor \neg p))$ negation & commutative

$\equiv (p \lor q) \land (\neg q \lor \neg p)$ identity law

$\equiv (\neg p \to q) \land (q \to \neg p)$ logical eq (1), double negation

$\equiv \neg p \leftrightarrow q$ Hence, verified logical eq(2)

Date:

QUESTION # 26

Logical Equivalence

$$p \to q \equiv \neg p \lor q \quad (1)$$

Distributive Laws

$$p \lor (q \land r) \equiv (p \lor q) \land (p \lor r)$$
$$p \land (q \lor r) \equiv (p \land q) \lor (p \land r)$$

$$(p \to q) \land (p \to r) \equiv p \to (q \land r)$$

Take $p \to (q \land r)$ logical equivalence (1)

$\equiv \neg p \lor (q \land r)$ distributive law

$\equiv (\neg p \lor q) \land (\neg p \lor r)$ logical equivalence (1)

$\equiv (p \to q) \land (p \to r)$

Thus, proved that $(p \to q) \land (p \to r)$ is logically equivalent to $p \to (q \land r)$

$$p \to (q \land r) \equiv (p \to q) \land (p \land r)$$

Verified

Date:

QUESTION #35

$(p \to q) \to r$ and $p \to (q \to r)$

p	q	r	$p \to q$	$q \to r$	$(p\to q)\to r$	$p \to (q\to r)$
T	T	T	T	T	T	T
T	T	F	T	F	F	F
T	F	T	F	T	T	T
T	F	F	F	T	T	T
F	T	T	T	T	T	T
F	T	F	T	F	F	T
F	F	T	T	T	T	T
F	F	F	T	T	F	T

Hence, it is proved that $(p \to q) \to r$ is not logically equivalent to $p \to (q \to r)$ because the last two columns of truth table do not contain same truth values in each row.

Date:

QUESTION # 36

p	q	r	$p \wedge q$	$p \to r$	$q \to r$	$(p \wedge q) \to r$	$(p \to r) \wedge (q \to r)$
T	T	T	T	T	T	T	T
T	T	F	T	F	F	F	F
T	F	T	F	T	T	T	T
T	F	F	F	F	T	T	F
F	T	T	F	T	T	T	T
F	T	F	F	T	F	T	F
F	F	T	F	T	T	T	T
F	F	F	F	T	T	T	T

Hence, proved that $(p \wedge q) \to r$ is not logically equivalent to $(p \to r) \wedge (q \to r)$ because the last two columns of truth table do not contain same truth values in each row.

QUESTION # 44

It is the conjunction. $(\wedge)$. The answer to the question is $p \wedge q \wedge \neg r$.

Where $p \wedge q \wedge \neg r$ is only true when $p, q, \neg r$ are true and p and q are true and r is false

QUESTION # 45

There are three ways in which exactly two of p, q and r can be true.

$p \wedge q \wedge \neg r$

$p \wedge \neg q \wedge r$

$\neg p \wedge q \wedge r$

Join them with disjunction ($\vee$) to obtain:

$(p \wedge q \wedge \neg r) \vee (p \wedge \neg q \wedge r) \vee (\neg p \wedge q \wedge r)$.

This is the required answer to the question.

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QUESTION # 4

$P(x,y)$ = "Student x has taken class y"

The domain of x = All students in your class

The domain of y = All computer science courses at your school

$\exists x P(x)$:
There exist an element x in the domain such that $P(x)$

$\forall x P(x)$:
$P(x)$ for all values of x in the domain.

a &

There is a student in your class that has taken a computer science course.

b &

There is a student in your class that has taken all computer science courses.

Date:

c)

All the students in your class have taken a computer science course.

d)

There is a computer science course that every student in your class has taken

e)

Every computer science course has been taken by some students in your class.

f)

All the students in your class have taken all the computer science courses.

QUESTION # 8

a)

"There is" means $\exists$

$\exists x \exists y\, Q(x,y)$

Date:

b $\mathcal{B}$–

No student includes every student

"Every" means $\forall$

$\forall x \forall y \neg Q(x, y)$

c $\mathcal{B}$–

"There is" means $\exists$ and "AND" means $\wedge$

$\exists x\,(Q(x, \text{Jeopardy}) \wedge Q(x, \text{Wheel of Fortune}))$

d $\mathcal{B}$–

"Every" means $\forall$

$\exists y \forall x\, Q(x, y)$

e $\mathcal{B}$–

At least two students means there exist two different students

$\exists x \exists z\,((x \neq z) \wedge Q(x, \text{Jeopardy}) \wedge Q(z, \text{Jeopardy}))$

explanation

$\exists x \exists z$: "there exist an x" and "there exist a z"

$(x \neq z)$: asserts that x and z are different individuals.

$Q(x, \text{Jeopardy})$: that x participates on Jeopardy.

$Q(z, \text{Jeopardy})$: that z also participates on Jeopardy.

Date:

QUESTION #9

$L(x,y)$ = " x loves y "

The domain of x = All people in the world.

The domain of y = All people in the world.

Negation $\neg p$: not p

Conjunction $p \vee q$: p or q

Disjunction $p \wedge q$: p and q

Conditional $p \rightarrow q$: if p then q

Biconditional $p \leftrightarrow q$: p if and only if q

Existential $\exists x\, P(x)$: There exist an element x in the domain such that $P(x)$.

Universal $\forall x\, P(x)$: $P(x)$ for all values of x in the domain.

a B

"Everybody" means "All people in the world"

$\forall x\, L(x, \text{Jerry})$

Date:

b Ans

"Everybody" means "All people in the world"

"Somebody" means "There exist a person in the world."

$\forall x \exists y (L(x,y))$

c Ans

Everybody refers to 'x' and somebody refers to 'y'

$\exists y \forall x L(x,y)$

d Ans

"Nobody" means "There does not exist a person in the world."

$\neg \exists x \forall y L(x,y)$

This is logically equivalent with $\forall x \exists y \neg L(x,y)$ by DeMorgan's Law of quantifiers.

e Ans

We could rewrite the given statement as "Lydia does not love somebody". "Somebody" means "There exist a person in the world."

$\exists y \neg L(\text{Lydia}, y)$

Date:

f B

We could rewrite the statement as "There is somebody whom everybody does not love."

$$\exists y \forall x \neg L(x,y)$$

g B

$$\exists x(\forall y\, L(y,x) \land \forall z\, (\forall y\, L(y,z) \rightarrow z = x))$$

explanation

$\exists x$: There exist a person x

$\forall y\, L(y,x)$: For all people y, y loves x

$\forall z\, (\forall y\, L(y,z) \rightarrow z = x)$: If there is any person z such that everybody loves z, then z must be the same person as x.

We could ~~rewe~~ rewrite the given sentence as "There is somebody x whom everyone loves and all people that are loved by everyone, then this person has to be x."

h B

We could rewrite the sentence as "There are two people y and z that Lynn loves and these two people are different and for all people that Lynn loves, these people then have to be either y or z."

Date:

$\exists y \exists z (L(Lynn, y) \land L(Lynn, z) \land y \neq z \land \forall w$

$(L(Lynn, w) \rightarrow (w = y \lor w = z)))$

i ẞ~

$\forall x \, L(x, x)$

Everyperson x loves x (himself / herself)

j ẞ~

We could rewrite the sentence as " There is a person x who loves y if and only if y is x (himself / herself)"

$\exists x \forall y (L(x, y)) \leftrightarrow x = y)$

QUESTION #10

$F(x, y) =$ " x can fool y "

The domain of x = All people in the world

The domain of y = All people in the world

Date:

a $\mathcal{B}$

$\forall x \, F(x, \text{Fred})$

b $\mathcal{B}$

$\forall y \, F(\text{Evelyn}, y)$

c $\mathcal{B}$

$\forall x \, \exists y \, F(x,y)$

"Everybody" means "All people in the world"
"Somebody" means "The exist a person in the world."

d $\mathcal{B}$

"Noone" means "There does not exist a person in the world."

$\neg \exists x \forall y \, F(x,y)$; Logically equivalent to $\forall x \, \exists y \, \neg F(x,y)$ by De Morgan

e $\mathcal{B}$

Everybody refers to y while somebody refers to x

$\forall y \, \exists x \, F(x,y)$

Date:

f:

"No one" means "There does not exist a person in the world"

$$\neg \exists x \, (F(x, Fred) \wedge F(x, Jerry))$$

g:

We could rewrite the sentence as "Nancy could fool two people y and z, and y and z can't be the same person all people that Nancy can fool then have to be either y or z"

$$\exists y \exists z (F(Nancy, y) \wedge F(Nancy, z) \wedge y \neq z \wedge \forall w \, (F(Nancy, w) \rightarrow (w = y \vee w = z))$$

h:

We could rewrite the sentence as "There is a person y whom everybody can fool and all other people whom everybody can fool them then have to be this person y."

$$\exists y (\forall x \, F(x, y) \wedge \forall z (\forall w \, F(w, z)) \rightarrow z = y)$$

Date:

i)

We could rewrite as ' Every person x can't fool x '

$\forall x \neg F(x,x)$

This is logically equivalent to $\neg \exists x F(x,x)$ by De Morgan's Law for quantifiers.

j)

We could rewrite as " There is a person x for whom there exists a person y that x can fool and all other people z that x can fool, z has to be y or z has to be x (himself /herself)

$$\exists x \exists y (L(x,y) \wedge \forall x (L(x,z) \rightarrow z = y \vee (z = x))$$

QUESTION # 14

$A(x,y)$ means " Student x speaks y "

Negation $\neg p$: not p

Date:

Conjunction $p \wedge q$: p and q

Disjunction $p \vee q$: p or q

Conditional $p \rightarrow q$: if p then q

Biconditional $p \leftrightarrow q$: p if and only if q

Existential $\exists x\, P(x)$: There exist an element x in the domain such that P(x)

Universal $\forall x\, P(x)$: P(x) for all values of x in the domain

a &

$\exists x\, A(x, \text{Hindi})$

b &

Let B(x,y) means "student x plays sports y"

$\forall x\, \exists y\, B(x,y)$

c &

Let C(x,y) means " Student x has visited y"

$\exists x\, [C(x, \text{Alaska}) \wedge \neg C(x, \text{Hawaii})]$

d &

Let D(x,y) means " Student x has learned programming language y "

Date:

$\therefore$

$\forall x \exists y D(x,y)$

e :

Let $E(x,y)$ mean " Student x has taken all courses offered by department y "

$\exists x \exists y E(x,y)$

F :

Let $F(x,y)$ means " Student x grew up in the same town as y "

$\exists x \exists y (x \neq y \land F(x,y) \land \forall z (F(x,z) \rightarrow (x = z \lor y = z)))$

g :

Let $G(x,y,z)$ means " student x has chatted with student y in chat group z. "

$\forall x \exists y \exists z G(x,y,z)$

Date: ______

QUESTION # 18

a)

Let $A(x,y)$ mean " consule x is accesible during fault conditiony."

$\forall y \exists x A(x,y)$

b)

Let $B(x,y)$ mean " e-mail address of user x has sent message y on the system that is contained in the archive"

and $C(x)$ mean " email address of user x can be retrieved."

$\forall x \exists y [B(x,y) \rightarrow C(x)]$

c)

Let $D(x,y)$ mean " mechanism x can detect security breach y"

and

$E(z)$ means " process z has been compromie"

$\forall y \exists x D(x,y) \leftrightarrow \exists z E(z)$

Date:

d $\mathscr{B}$

Let $F(x,y,z)$ means "path x connects distinct end-points y and z"

$\forall y \forall z \exists x \exists a [x \neq a \rightarrow (F(x,y,z) \wedge F(a,y,z))]$

e $\mathscr{B}$

Let $G(x,y)$ means "x knows password of user y"

and

$H(x)$ mean "x is a system administrator"

$\underbrace{\forall x [H(x) \rightarrow \forall y\, G(x,y)]}_{1^{st} \text{ part}} \wedge \underbrace{\neg \exists x [\neg H(x) \wedge \forall y\, G(x,y)]}_{2nd \text{ part}}$

explanation

1st part means; that any one who is system admin knows the password of every user.

2nd part means; that no non-admin user knows the password of every user y.

QUESTION # 36

Negation $\neg p$: not p

Conjunction $p \wedge q$: p and q

Disjunction $p \vee q$: p or q

Conditional $p \rightarrow q$: if p then q

Biconditional $p \leftrightarrow q$: p if and only if q

Existential $\exists x\, P(x)$: There exist an element x in the domain such that $P(x)$

Universal $\forall x\, P(x)$: $P(x)$ for all values of x in the domain

Logical Equivalence $\neg(p \rightarrow q) \equiv p \wedge \neg q$ (1)

Double negation law $\neg(\neg p) \equiv p$

De Morgans's Law $\neg(p \wedge q) \equiv \neg p \vee \neg q$
$\neg(p \vee q) \equiv \neg p \wedge \neg q$

De Morgan's law for quantifiers $\neg \exists x\, P(x) \equiv \forall x \neg P(x)$
$\neg \forall x\, P(x) \equiv \exists x \neg P(x)$

Date:

2 B

Let $A(x)$ mean " x has lost more than one thousand dollars playing in the lottery."

We can rewrite the given sentence as " There does not exist a person that has lost more than one thousand dollars palaying in the lottery."

$$\neg \exists x A(x)$$

We are interested in the negation of this statement. Use double negation law:

$$\neg [\neg \exists x A(x)]$$

$$\equiv \exists x A(x)$$

Translating the sentence back to English:

There exist a person that has lost more than one thousand dollars playing in the lottery.

Date:

b 8

Let $B(x,y)$ means:
" x has chatted with y"

and the domain is all students of this class

We can rewrite the given sentence as:
" There is a student in this class who has chatted with one student and this student is not himself / herself and for all people the student chatted with, this student then has to be himself or the one student he chatted with.

$$\exists x \exists y [B(x,y) \land x \neq y \land \forall z (B(x,z) \to (z = x \lor z = y))]$$

We are interested in the negation:

$$\neg \exists x \exists y [B(x,y) \land x \neq y \land \forall z (B(x,z) \to (z = x \lor z = y))]$$

Use De Morgan's Law for quantifiers (twice):

$$\equiv \forall x \neg \exists y [B(x,y) \land x \neq y \land \forall z (B(x,z) \to (z = x \lor z = y))]$$

$$\equiv \forall x \forall y \neg [B(x,y) \land x \neq y \land \forall z (B(x,z) \to (z = x \lor z = y))]$$

Use De Morgan's Law

$$\equiv \forall x \forall y [\neg B(x,y) \lor \neg (x \neq y) \lor \neg \forall z (B(x,z) \to (z = x \lor z = y))]$$

Date:

Use De Morgan's Law for quantifiers:

$\equiv \forall x \forall y [\neg B(x,y) \vee x = y \vee \exists z \neg (B(x,z) \wedge \neg (z = x \vee z = y))]$

Use logical equivalence (1):

$\equiv \forall x \forall y [\neg B(x,y) \vee x = y \vee \exists z (B(x,z) \wedge \neg (z = x \vee z = y))]$

Use De Morgan's Law:

$\equiv \forall x \forall y [\neg B(x,y) \vee x = y \vee \exists z (B(x,z) \wedge (\neg (z = x) \wedge \neg (z = y)))]$

$\equiv \forall x \forall y [\neg B(x,y) \vee x = y \vee \exists z (B(x,z) \wedge (z \neq x) \wedge (z \neq y)]$

Translating the sentence back to English:

All students have chatted with no one or has chatted with more than 2 people.

c B

Let $C(x,y)$ mean:
" x has sent email to y"

and the domain is all students of this class.

Date:

We can rewrite the given sentence as:
" There is not a student in this class who has sent an e-mail to one person and to another person, and these two person, and these two people are different and these people are different from the original person and did not send an e-mail to anybody else.

$$\neg \exists x \exists y \exists z [C(x,y) \wedge C(x,z) \wedge x \neq y \wedge x \neq z \wedge y \neq z$$
$$\wedge \forall w (C(x,w) \to (w = x \vee w = y \vee w = z))]$$

We are interested in negation:

$$\neg(\neg \exists x \exists y \exists z [C(x,y) \wedge C(x,z) \wedge x \neq y \wedge x \neq z \wedge y \neq z$$
$$\wedge \forall w (C(x,w) \to (w = x \vee w = y \vee = w = z))]$$

Use the double negation law:

$$\exists x \exists y \exists z [C(x,y) \wedge C(x,z) \wedge x \neq y \wedge x \neq z \wedge y \neq z$$
$$\wedge \forall w (C(x,w) \to (w = x \vee w = y \vee w = z))]$$

Translating the sentence in English

There is a student in this class that has sent e-mail to exactly to other students in this class.

Date:

d)

Let $D(x,y)$ mean:
"student x has solved exercise y in this book."

We can rewrite the given sentence as:
"There exist a student that has solved all exercises in this book."

$\exists x \forall y\, D(x,y)$

The negation is:

$\neg \exists x \forall y\, D(x,y)$

Use De Morgan's Law for quantifiers (twice)

$\equiv \forall x \neg \forall y\, D(x,y)$

$\equiv \forall x \exists y \neg D(x,y)$

Translating the sentence back to English

For all students there exist an exercise in this book that the student did not solve

e)

Let $E(x,y,z)$ be
"student x has solved exercise y in section z of this book."

$\neg \exists x \exists y \forall z\, E(x,y,z)$

The negation of the sentence is:

$\equiv \neg(\neg \exists x \exists y \forall z\, E(x,y,z))$

Using double negation law:

$\equiv \exists x \exists y \forall z\, E(x,y,z)$

Translating into English:

There is a student that solved at least one exercise of every section of this book.

QUESTION #12

$I(x)$ = x has internet connection

$C(x,y)$ = x and y have chatted over internet

Domain of x and y = All students in your class.

a

$\neg I(\text{Jerry})$

b

$\neg C(\text{Rachael}, \text{Chelsea})$

c

$\neg C(\text{Jan}, \text{sharon}) \land \neg C(\text{Sharon}, \text{Jan})$

d

$\neg \exists x\, C(x, \text{Bob})$

e

$\forall x\, [(x \neq \text{Joseph}) \rightarrow C(\text{Sanjay}, x)]$

f

$\exists x\, \neg I(x)$

g

$\neg \forall x\, I(x)$

h

$\exists!$ means exactly one

$\exists! x\, I(x)$

i Ans

$\exists x\, [\neg I(x) \wedge \forall z\, (x \neq z\ I(z))]$

j Ans

$\forall x\, [I(x) \rightarrow \exists y\, C(x,y)]$

K Ans

$\exists x\, [I(x) \rightarrow \forall y\, \neg C(x,y)]$

l Ans

$\exists x \exists y\, [x \neq y \rightarrow \neg C(x,y)]$

m Ans

$\exists x \forall y\, [C(x,y)]$

n Ans

$\exists x \exists z\, [x \neq z \wedge \forall y\, (\neg C(x,y) \wedge \neg C(z,y))]$

o Ans

$\exists x \exists z\, [x \neq z \rightarrow \forall y\, (C(x,y) \vee C(z,y))]$

Date:

QUESTION

Convert $(180)_{10}$ into Binary

2	180	
2	90	0
2	45	0
2	22	1
2	11	0
2	5	1
2	2	1
	1	0

$(180)_{10} \rightarrow (10110100)_2$

Convert $(56)_{10}$ into Binary

2	56	
2	28	0
2	14	0
2	7	0
2	3	1
	1	1

$(56)_{10} \rightarrow (111000)_2$

Date:

QUESTION

Convert the binary number $(0101001)_2$ into a decimal system.

$= (0 \times 2^6) + (1 \times 2^5) + (0 \times 2^4) + (1 \times 2^3)$

$+ (0 \times 2^2) + (0 \times 2^1) + (1 \times 2^0)$

$= 0 + 32 + 0 + 8 + 0 + 0 + 1$

$= 41$

$(0101001)_2 \rightarrow (41)_{10}$

Convert the given binary number $(100001)_2$ into a decimal number system.

$= (1 \times 2^5) + (0 \times 2^4) + (0 \times 2^3) + (0 \times 2^2)$

$+ (0 \times 2^1) + (1 \times 2^0)$

$= 32 + 0 + 0 + 0 + 0 + 1$

$= 33$

$(100001)_2 \rightarrow (33)_{10}$

Date:

QUESTION

Subtract $(1010)_2$ $(1010)_2$ from $(1111)_2$ $(1111)_2$ using 2's complement.

```
        1 0 1 0              1 1 1 1
      x 1 0 1 0            x 1 1 1 1
      ---------            ---------
        0 0 0 0            1 1 1 1 1
      1 0 1 0 x          1 1 1 1 1 x
    0 0 0 0 x x        1 1 1 1 x x
 + 1 0 1 0 x x x    + 1 1 1 1 x x x
 ---------------    ---------------
  1 1 0 0 1 0 0      1 1 1 0 0 0 0 1
```

$(11100001)_2 - (1100100)_2$

Subtrahend is $(1100100)_2$

2's complement of subtrahend ; 0011011

now add 1 ; 0011011 + 1

→ 0011100

now, adding in miniuend;

```
          1 1 1 0 0 0 0 1
        +   0 0 1 1 1 0 0
carry bit (1) 1 1 1 1 1 0 1
```

Date

Since, there is a carry bit, so dropped this carry bit 1

The result is $(111101)_2$ Therefore,
$(111)_2 (111)_2 - (1010)_2 (1010)_2 = (111101)_2$

Subtract $(1010)_2 (1010)_2$ from $(1000)_2 (1000)_2$ using 2's complement method

```
          1 0 1 0
        x 1 0 1 0
        ---------
          0 0 0 0
        1 0 1 0 x
      0 0 0 0 x x
   + 1 0 1 0 x x x
   ---------------
   1 1 0 0 1 0 0
```

```
          1 0 0 0
        x 1 0 0 0
        ---------
          0 0 0 0
        0 0 0 0 x
      0 0 0 0 x x
   + 1 0 0 0 x x x
   ---------------
   1 0 0 0 0 0 0
```

To find $(1000000)_2 - (1100100)_2$ we find 2's complement of subtrahend which will be 0011011, now add 1 in it

i.e.

```
  0011011
       +1
  -------
  0011100
```

now add it to minivend

```
    1 0 0 0 0 0 0
  + 0 0 1 1 1 0 0
  ---------------
    1 0 1 1 1 0 0
```

Date:

Since there is no carry bit so taking 2's complement of the result

0100011 ; now adding 1

⇒ 0100011 + 1 = 100100

So the answer is $(100100)_2$

QUESTION

Add the following binary numbers

i B~

			1	1	1	1	0	0	0	0	
+	1	0	0	0	0	1	1	0	0	1	
	1	1	0	0	0	0	1	0	0	1	

The answer is $(1100001001)_2$

ii B~

		1	0	1	0	1	0	0	0
+	1	0	0	0	1	0	0	1	
1	0	0	1	1	0	0	0	1	

The answer is $(100110001)_2$

Date:

QUESTION

Perform $(1010)_2 - (101111)_2$ using 1's and 2's complement.

Subtracting by using 1's Complement

Finding 1's complement of subtrahend i.e $(010000)_2$

Adding it with miniuend:

i.e
$$\begin{array}{r} 010000 \\ +\ 1010 \\ \hline 011010 \end{array}$$

Since there is no carry bit 1, we calculate 1's complement of (011010) we get (100101) and putting -ve sign

Hence, the result is $-(100101)_2$

Subtracting Using 2's Complement Method

First finding 2's complement of subtrahend i.e $(101111)_2 \rightarrow (010000)$

now adding +1 in it. (LSB)
$(010000) + (1) = (010001)$

Date:

now, add with minuend

$$\begin{array}{r} 010001 \\ +\ 1010 \\ \hline 011011 \end{array}$$

Since, there is no carry bit over 1, we take 2's complement of the result.

i.e (100100)

now add (1) in LSB

$$\begin{array}{r} 100100 \\ +\ 1 \\ \hline 100101 \end{array}$$

place -ve sign - $(100101)_2$

So the result is - $(100101)_2$

Subtract (1010) from (1111) using 1's and 2's complement method

Given question is (1111) - (1010)

Using 1's Complement

first calculating 1's complement of (1010) is (0101)
Adding (0101) with miniuend (1111) we get

$$\begin{array}{cccccc} & {}^{1}0 & 1 & {}^{1}0 & 1 \\ + & 1 & 1 & 1 & 1 \\ \hline \boxed{1} & 0 & 1 & 0 & 0 \end{array}$$

Since there is carry bit over 1, removing that extra bit and ting on it.

$$\begin{array}{ccccc} & 0 & 1 & 0 & 0 \\ & & & + & 1 \\ \hline & 0 & 1 & 0 & 1 \end{array}$$

Hence, the result is $(0101)_2$

Using 2's Complement

First calculating 2's complement of subtrahend (1010) is $(0101)_2$. now adding (1) to LSB.

$(0101) + (1) = (0110)$

now adding it with minivend

$$\begin{array}{cccccc} & {}^{1}0 & {}^{1}1 & 1 & 0 \\ + & 1 & 1 & 1 & 1 \\ \hline \boxed{1} & 0 & 1 & 0 & 1 \end{array}$$

Since there is a carry bit over 1. drop it, we get $(0101)_2$

Hence, the result is $(0101)_2$